Week 4 Part E — written justification

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The four panels and their questions

1. Daily sessions: how is traffic moving once the weekday cycle is separated? A time-series line uses position for sequence; the faint raw daily series remains under a named 7-day centred mean. The mean is retrospective; the live counterpart must be trailing and disclose its lag.
2. Weekday × hour heatmap: when is traffic typically concentrated? A sequential blue scale encodes an ordered mean; masked missing cells are not filled with zero.
3. Treemap: where is the net-revenue mass across category → sub-category? Area is appropriate for hierarchy overview, each parent category has a consistent identity hue, and values are printed because area ranks below position. Negative rows are shown as a separate return-credit issue, never as area.
4. Sorted bars: how much net revenue does each sub-category contribute? Common-baseline length gives more precise comparison than area or angle. No Other bucket was used: 0 sub-categories / PKR 0 are hidden; the cut is at the eight sub-category level.

Time, parse and aggregation

The timestamp format is explicit by logger: 9,019 ISO strings use %Y-%m-%d %H:%M:%S; 3,651 slash strings use day-first %d/%m/%Y %H:%M. I checked the parsed min/max rather than trusting mixed-format success. The sorted, de-duplicated hourly index has 312 missing hours; -1 is an invalid sessions sentinel. For daily flows, valid hourly session totals are divided by observed hours and multiplied by 24; this estimates a full-day total and assumes missing hours are not systematically different. Missing whole days remain NaN and form line breaks; the final 11-day September 2025 tail is omitted as incomplete. The heatmap uses mean observed sessions per hour. Daily totals use sum semantics (with transparent coverage correction), never a sum of an instantaneous level.

The 7-day window was fixed before inspection because the cycle it removes is weekly. The chart keeps raw values and names the window. A 28-day version and an exhaustive 12-start × 5-window slope scan are in the appendix. The annual decomposition is not identifiable from 518 days under seasonal_decompose's two-cycle requirement; it is reported as unavailable instead of extrapolated.

Row and hierarchy accounting

- Duplicate timestamp rows removed: 286; all duplicate groups agree exactly on measures and normalized device.
- Invalid sessions=-1 observations: 287 values excluded as missing.
- Missing hourly periods from the complete expected index: 312; this includes 13 dates with no hourly data.
- Incomplete final month: 11 recorded September days removed from the complete-month analysis.
- Duplicate SKU rows removed: 1. Negative net-revenue SKU rows excluded from area: 7, net -1.0693 PKR m, retained as an explicit negative audit. Positive subcategory hierarchy total: 1,712.4508 PKR m. The cut is category → sub-category; all eight sub-categories remain visible, so Other is empty.

The same fixed window and start date would be used if the result weakened the story. partB reports both rising and falling combinations, making the selection sensitivity visible rather than hiding rejected views.